

Supplementary Material: Local Refinement with Frozen Estimator Applications

Internal draft for author review

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1 Status and reading guide

Tables and figures copied from the original manuscript describe historical YOLO experiments only. The DOPE extension is identified separately below; its unmeasured values are never filled from that panel. This supplement preserves the distinction between completed development evidence and independent confirmation. The machine-readable status is in `FINAL_STATUS.json` under the submission experiment directory. All three matched-budget fits were completed; the learning curves are diagnostic, not a best-checkpoint selector. Independent real-session confirmation is not yet available. No method’s incomplete result is represented by another model or a zero-valued measurement.

2 Historical YOLO development results

Original-pixel medians and P90 are pooled over each model’s supervised matched points. Frame mean is the mean of matched per-frame errors, not the pooled median. Gross20 and PCK are percentages; PCK retains the full 2,818-point ground-truth denominator. Raw prior rows, when available, precede no claim that the selected capped wrapper is the original standalone method.

Model	Med	P90	Mean	Gross20	PCK5	PCK10	PCK20	N
R0	6.616	38.670	21.673	17.20	37.19	63.73	80.98	2756
P1	5.851	38.025	20.930	15.57	42.26	67.64	82.58	2756
P2	5.943	37.728	21.010	15.89	42.16	67.07	82.26	2756
P3	5.921	37.546	20.949	15.42	41.66	67.32	82.72	2756
L1	6.052	37.204	21.145	16.07	41.23	66.57	82.08	2756
L2	6.037	37.302	21.085	15.93	41.06	66.93	82.22	2756
L3	6.121	37.453	21.164	16.00	41.02	66.86	82.15	2756
D1	6.366	37.479	21.411	16.11	39.14	64.66	82.04	2756
D2	6.457	37.810	21.456	16.44	38.75	64.62	81.72	2756
D3	6.467	37.410	21.435	16.22	38.96	64.44	81.94	2756
PRIOR1	5.569	37.218	20.784	15.49	44.78	68.31	82.65	2756
PRIOR1_raw	5.569	36.776	20.655	15.42	44.78	68.28	82.72	2756
PRIOR2	5.541	37.373	20.738	15.64	44.75	68.84	82.51	2756
PRIOR2_raw	5.541	37.295	20.594	15.53	44.78	68.88	82.61	2756
PRIOR3	5.596	37.089	20.830	15.71	43.90	68.06	82.43	2756
PRIOR3_raw	5.599	36.721	20.700	15.60	43.90	68.10	82.54	2756

3 Geometry-reference results

Rotation and yaw are degrees; translation is centimeters. These are not independent physical 6D measurements. Pose coverage is separate from box matching: the canonical prediction-only selector may return a pose from a top detection that does not meet the 2D matching gate. No new symmetry minimization was introduced to excuse near-square axis confusion.

Model	Rot(deg)	Yaw(deg)	Trans(cm)	IoU3D	ADD AUC	Coverage
R0	2.262	1.231	7.897	0.603	0.428	1.000
P1	2.089	1.074	6.865	0.642	0.454	1.000
P2	2.106	1.136	6.917	0.637	0.456	1.000
P3	2.073	1.156	7.678	0.637	0.454	1.000
L1	2.110	1.115	7.609	0.637	0.450	1.000
L2	2.103	1.122	7.364	0.636	0.452	1.000
L3	2.136	1.132	7.672	0.625	0.445	1.000
D1	2.201	1.192	7.570	0.621	0.441	1.000
D2	2.181	1.182	7.540	0.619	0.442	1.000
D3	2.188	1.192	7.682	0.616	0.440	1.000
PRIOR1	2.018	1.100	7.185	0.614	0.455	1.000
PRIOR1_raw	2.018	1.100	7.185	0.616	0.456	1.000
PRIOR2	2.004	1.102	6.675	0.628	0.467	1.000
PRIOR2_raw	2.010	1.105	6.844	0.628	0.464	1.000
PRIOR3	2.132	1.190	6.825	0.619	0.460	1.000
PRIOR3_raw	2.132	1.190	6.825	0.619	0.460	1.000

4 Registered geometry and camera inputs

Object	Long(m)	Short(m)	Height(m)	Frames
Plastic	1.30	1.10	0.11	194
Wood	0.80	0.59	0.14	125

Width	Height	f_x	f_y	c_x	c_y	Frames
640	480	605.906	605.970	317.596	256.292	28
640	480	608.298	607.846	326.401	239.547	146
640	480	614.184	614.313	329.280	234.529	100
1280	720	908.860	908.955	636.394	384.438	45

Intrinsics above are annotation records in pixels, not newly measured calibration accuracy. The canonical solver passes no distortion coefficients. The records inspected here provide resolution and intrinsics, not an independently verified lens/calibration uncertainty report. Object dimensions are the unchanged registered geometry; the evaluation contract does not claim new physical inspection. The declared long-axis equivalence allows a half turn, but not a quarter turn. The wood object is not established as visually symmetric under the half turn. These benchmark conventions do not certify interchangeable physical fork-entry geometry.

5 Complete timing summaries

The expanded panel was measured in the same session for all comparators.

All values are milliseconds. The same-session expanded panel is used after the prior evaluation is complete; otherwise only the explicitly historical timing is available. Every measured repeat is retained. Model and image loading are excluded, but actual RGB preprocessing, detector forward, feature sampling/refinement, restoration and canonical PnP are included in their named intervals. Allocated-memory figures do not include unrelated desktop processes.

Model	2D med	mean	P90	Total med	mean	P90	Add2D med
R0	9.733	10.017	11.803	11.217	11.522	13.298	0.000
P1	13.661	14.031	15.730	15.352	15.496	17.096	4.056
P2	13.622	13.996	15.886	15.280	15.465	17.332	3.802
P3	13.784	14.096	15.984	15.550	15.596	17.355	4.077
L1	18.226	18.106	20.105	19.736	19.598	21.496	7.975
L2	18.150	18.163	20.161	19.858	19.670	21.572	8.115

Model	2D med	mean	P90	Total med	mean	P90	Add2D med
L3	18.178	18.194	20.097	19.734	19.685	21.457	8.194
D1	13.231	13.509	15.482	14.800	15.022	16.941	3.343
D2	12.979	13.464	15.558	14.800	14.993	16.948	3.284
D3	13.226	13.444	15.365	14.773	14.927	16.832	3.211
PRIOR1	27.537	27.380	28.439	29.020	28.926	29.890	17.466
PRIOR2	27.844	27.824	28.737	29.320	29.308	30.179	17.937
PRIOR3	27.844	27.653	28.594	29.297	29.123	29.994	17.918

The first complete balanced panel supplies every latency sample. An earlier incomplete attempt failed an overly strict bit-exact replay check and is retained separately, not selected or rejected by speed. Fixed-input repeated GPU forwards located the prior’s first numerical differences in its first upsampling stage; the stem and ResNet blocks were identical in that diagnostic. The frozen nondeterministic cuDNN settings and all model weights remain unchanged. Corrected prior coordinates are checked in crop space using the existing network validation’s absolute component, 3×10^{-4} pixels, with zero relative allowance. Detector contracts and center8 remain bit-exact. Historical coordinates reproduce canonical pose metrics; current prior coordinates must preserve hypothesis and coverage. The final DEV accuracy tables still use the original frozen predictions.

After all balanced timing samples passed, the model-release guard prevented reuse of the same process for isolated memory measurements. The saved latency samples were retained unchanged; allocation was measured in a new process for each model. A further untimed coordinate replay audit belongs to that memory phase, not to the earlier timing samples. Per-sample timed numerical bounds passed, but individual drift magnitudes and OpenCV/inter-operation thread observations were not saved before the guard stopped the process. Those missing observations are explicitly distinguished from the separately recorded memory-phase settings. Recovery records and both original attempts are retained; no fastest session is chosen.

6 Controlled training and convergence limits

Parameter, memory and cache accounting

Model	Learned	Total	Add/R0(%)	Inf.MiB	Train MiB	s/update
R0	0	2615142	0.00	190.01	NM	NM
P1	18962	2634104	0.73	190.09	NM	0.0590
D1	19450	2634592	0.74	190.09	NM	0.0662
L1	19810	2634952	0.76	190.09	NM	NM
PRIOR1	68661641	71276783	2625.54	454.84	7779.02	0.4163

The shared R0 has 2,615,142 parameters. The recorded first cache pass processed 60,000 records in 1668.83 seconds; bound cache arrays occupy 68.677 GiB. Head fit time excludes that extraction. NM indicates an unavailable historical allocation measurement, not zero cost. R0 is not newly trained. CPU: 11th Gen Intel(R) Core(TM) i9-11900 @ 2.50GHz. Timed code fixes Torch intra-operation threads to 4. Inter-operation and OpenCV thread observations were not durably captured in the timed phase. In the separate fresh-process memory phase, observed settings were Torch intra-operation 4, inter-operation 8, OpenCV 1 (R0 process; all models are in the runtime JSON). These are not retrospectively labeled timed-phase observations. P and D share source order, feature evidence and historical optimizer/exposure settings, but their loss and output geometry differ. D’s small fixed-probe decrease is neither a convergence guarantee nor evidence of fundamental regression inferiority. The prior uses ImageNet TF-Slim ResNet152 initialization and the official loss/Adam semantics adapted to nine pallet points. Its sample order is exactly the existing P order and its nominal batch is sixteen. Source train/cal probes at step0 and every500 updates are diagnostic only. No checkpoint or wrapper is selected on DEV.

7 Operator and weight provenance

The official PoseFix source commit is 5556364bb0f43b0743a5fcd820de48f34b3d4360. Its MIT notice and the TensorFlow authors’ Apache-2.0 attribution are retained. The official TF1 CPU reference runs in an isolated Python3.7

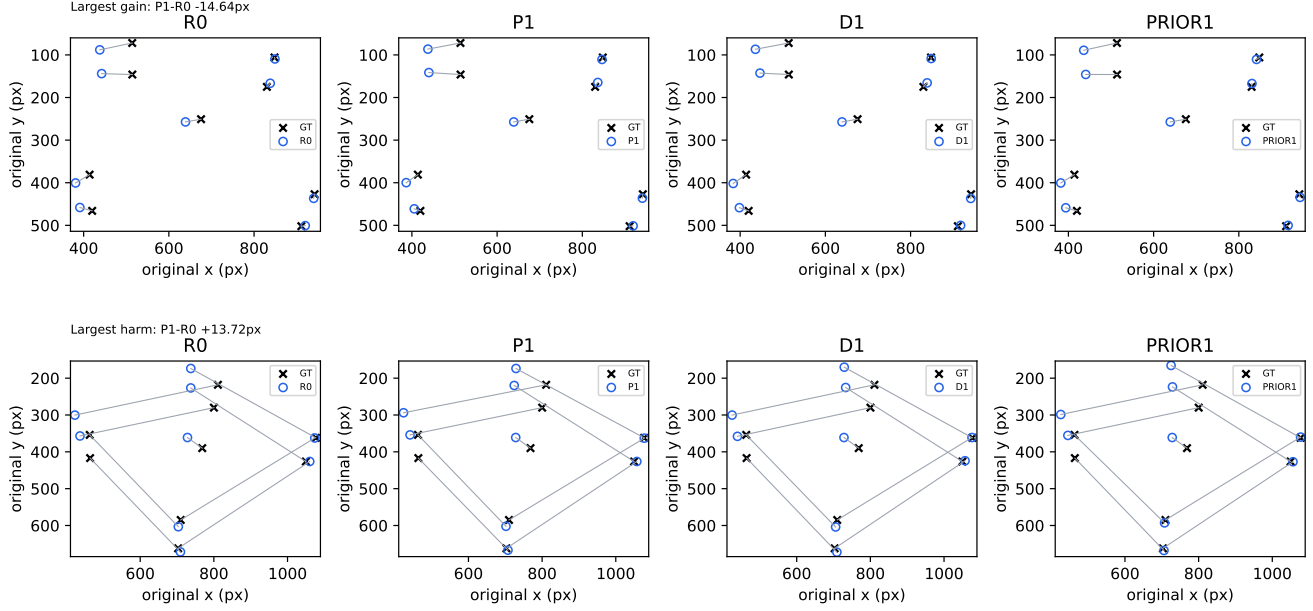


Figure 1: Actual improving and harming DEV cases: the lowest and highest per-frame median difference P1–R0, with lexical frame-ID ties. Same-frame GT and every plotted model use unchanged original coordinates and shared limits. Membership and raw sources are in `FIGURE_MANIFEST.json`. Images are not republished without verified image rights; these coordinate-only panels do not establish physical boundary correctness or independent generalization.

environment, and the PyTorch port uses the existing YOLO runtime without replacing that environment’s Torch, CUDA, or NumPy.

The reference initializes new weights using the official graph and transfers ImageNet RGB/backbone variables. Added pose channels remain at the official seeded initialization. The port maps convolution HWIO to OIHW, transpose convolution HWOI to IOHW, and every BN affine/moving variable. Missing and unexpected keys are explicitly checked. Full eval-coordinate and stage-feature comparisons use the actual source images and the same weights; scalar/loss-gradient and moving-statistics comparisons separately test training operators.

The graph’s nominal BN epsilon is not its effective value: the TF1 fused wrapper enforces 1.001×10^{-5} . The torch port matches this executed behavior. The Gaussian amplitude is 255 exactly once. Expectation coordinates are zero-based and scaled by 4 at the locked resolution. Bilinear target neighbors outside the grid are excluded and remaining mass normalized; nonfinite/outside-crop targets are masked in training, without altering evaluation membership. Raw RGB is explicitly the requested adaptation from the inspected BGR-preserving source loader.

The full batch 16 resource probe and actual parameter update are recorded separately from the three main fits. Main initialization and optimizer state do not reuse that smoke update. Checkpoints contain optimizer, Python/NumPy/CPU/CUDA RNG, step and exact order-prefix hash. The loader has no hidden random augmentation state; pending RGB reads can be reconstructed deterministically from the next recorded row.

8 Research history and interpretation

Prior Hough/line, active-adaptation, replay and robustness investigations motivated subsequent decisions, but do not become additional unseen validation sets. Old code, checkpoints, metrics and verdicts remain unchanged. This selection history means DEV is reused development evidence. The fixed model is not redesigned in response to an unfavorable prior comparison.

The baseline checkpoint is not purely synthetic-from-scratch: the recorded initialization checkpoint identifies COCO-pose training, despite a historical output directory containing the word scratch. P’s additional training uses synthetic labels only. ImageNet supervision for the added prior is separately disclosed. These distinctions prevent an overbroad claim that no real annotation contributed anywhere.

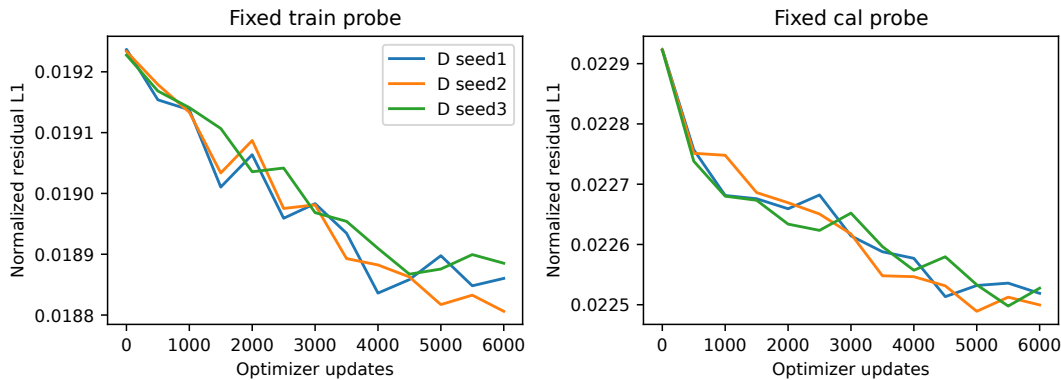


Figure 2: Already-completed D fixed probes. Their limited change is disclosed alongside the comparison, not suppressed. Loss magnitudes are not compared with P’s different objective.

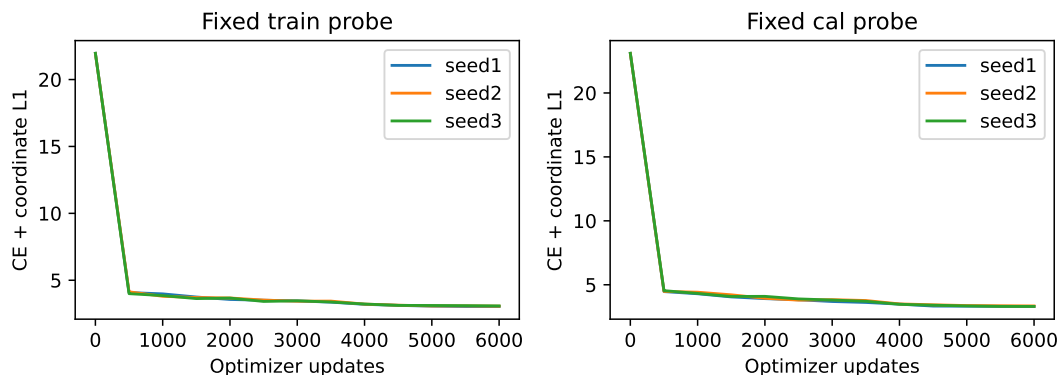


Figure 3: Prior fixed train/cal probes from the completed bounded fits. These curves did not select a checkpoint or extend training.

9 Independent data and annotation handoff

The actual collection and schema are in `HUMAN_INPUTS_REQUIRED.txt` and `CONFIRMATION_SCHEMA.json`. The starting plan is 12 separate sessions with 15 prespecified frames each; at least 30 frames receive two independent prediction-blinded annotations and blind adjudication. The count is not a power guarantee. Preserve original videos, timestamps, capture groups, camera/mount, calibration, actual object dimensions/identity and prior-use declarations. Changed image encoding or filename does not prove independence from an old scene.

Unreviewed provenance and an unfrozen comparator panel block confirmation only, not the completed development experiment. New-frame reference geometry must be bound to its camera/dimensions/index convention before a 6D claim. Human annotation, author identities, funding and consent cannot be fabricated by the CLI. These obligations are separate from compilation and code checks.

10 Three-estimator scope and outstanding evidence

The third application is specified as a full-image, nine-landmark SimpleBaseline-derived ResNet18 with ImageNet initialization. Its baseline training, frozen P/D head fits and measured outcomes remain pending in this revision. Completion of DOPE alone does not satisfy the three-estimator evidence requirement. The third-estimator implementation and receipt-backed status are described below. No accuracy or latency is inferred from an implemented interface or a CPU input check.

11 DOPE extension: frozen interface and training lineage

DOPE source training/evaluation results have not been inserted. No DOPE accuracy, coverage, pose benefit, parameter count or measured latency is claimed in this draft. The frozen checkpoint is `weights/backbone_dope_final_v1/run/final_net_epoch_0060.pth`; its SHA-256 begins `0de80490cb3b4f9b`, with the full digest in the protocol. The actual original training loop includes epochs 0 through 60. Original DOPE training uses random crop 400 and its recorded rotation, color, noise, and Torch FP32 normalization. The fixed inference adapter instead follows the existing `reflect100-resizeback-short400` path with its existing NumPy-to-FP32 normalization. A historical header value of 448 does not describe the executed crop. No equal-baseline-epoch or identical-training/inference-normalization claim is made.

The new adapter takes BGR input and returns explicit composed affine matrices, native/canvas/network sizes, semantic point validity, hull and confidence. Taps are post-ReLU 17/26; layer 16 alone is a convolution, not the feature used. Anisotropic inverse restoration adds the transformed displacement to the retained FP64 original point. This keeps a zero move exactly equal to the baseline and prevents a fresh FP32 reprojection from introducing approximately 10^{-5} -pixel roundoff. Fixed image-size, corner-index, GT-padding, camera and dimension contracts are required before scoring; there is no silent ID or geometry fallback.

The added P and D heads share 256/stride4 and 128/stride 8 evidence, 16-channel projections and 24 encoded channels. The backbone is in evaluation mode with gradients disabled. Newly initialized heads alone are optimized; an existing YOLO head is not a warm start. The source protocol fixes three seeds, 6,000 updates per head, batch 16 and six final fits. The AdamW learning rate is 10^{-3} , weight decay 10^{-4} , betas (0.9, 0.999), warmup 100, cosine terminal fraction 0.1, and gradient clip 5. These are head budgets, not convergence certificates for either family or equality of baseline-training cost.

Source prediction metadata is cached over the fixed source manifest; large full-resolution feature maps are not retained as a training cache. Online frozen VGG feature extraction therefore contributes actual training cost. Exact source image identities, usable rows, per-seed order hashes, final checkpoints, calibration temperatures and selected movement rules belong to the new receipts. The source mask can differ from the historical 55,915 usable YOLO rows. Nominal exposures must not be confused with successfully supervised corners; supported-frame loss reductions are retained.

12 DOPE results, missingness and paired uncertainty

DOPE per-seed table: pending. Seven rows (DOPE, D1–D3, P1–P3) are reserved; their complete denominator is 319 images and 2818 supervised landmarks. DOPE paired intervals: pending. The frozen session-bootstrap outputs will be inserted without dropping support mismatches or failed resamples. DOPE source evidence: pending. Actual usable rows, head parameter counts, completed updates, calibration temperatures, selected movement rules and heldout results require their own receipts. The seven fixed arms are baseline DOPE and D1–D3/P1–P3. The complete population contains 319 positive images, 13 sessions and 2818 supervised keypoints. Every arm retains every population row, including failed or incomplete predictions. The conditional nine-point statistic requires the original box-IoU match and a finite nine-point prediction. Full-GT PCK counts excluded predictions as failures; separately labeled partial-point PCK is diagnostic only. No additional 2689-negative DOPE AP result is implied.

The evaluator reuses the canonical camera and dimension mapping, prediction-only choice of width or depth, and SQPnP followed by LM refinement. It does not let the reference pose choose a prediction hypothesis. Fewer than six finite corners, selector exceptions and failed PnP produce explicit failed rows. Pose medians use available fits and are always accompanied by coverage with denominator 319. A changed support prevents a conditional paired confidence interval rather than silently selecting a favorable intersection. Reference rotation/translation remain geometry-derived, with no new physical metrology.

Paired session and frame bootstrap reuse the existing seed 20260914 and 10000 draws. The family statistic averages per-seed pooled statistics, not predictions, and compares with one baseline. Source-selected wrappers do not use real outcomes, but the decision to run this second-estimator study followed earlier research on the same DEV population. The intervals are therefore exploratory and unadjusted. No independent TEST exists in this extension.

13 DOPE runtime and publication records

DOPE cost measurements: pending. Historical YOLO latency is not an estimate of this path. The prescribed seed-1 timing panel must finish before reporting DOPE overhead. The new timing protocol records full DOPE+head RGB

computation; no cached intermediate substitutes for the measured inference path. Seed 1 represents deployment without selecting the fastest seed. Keypoint and PnP-inclusive timings remain separate, and unavailable poses stay visible. Different timing sessions are not described as the historical thirteen-model session. Actual runtime receipts are needed before making any overhead or total-cost claim for DOPE.

The new experiment scripts are under `scripts/research/pallet_dope_refiner_20261001_v1`; the numerical records are under the matching `_docs/experiments/pallet_dope_refiner_20261001_v1` directory. The manuscript update script checks input bindings, population, arms and explicit completion before inserting results. `DOPE_RESULTS_BINDINGS.json` identifies inserted evidence, and `HISTORICAL_ASSETS.json` binds copied original manuscript assets. Pending evidence is stated explicitly rather than replaced by historical YOLO measurements. Build receipts certify typesetting only, not independent experimental verification or author approval.

14 ResNet18 application: implementation and measured status

ResNet18 baseline completion and before/after measurements remain pending. The third estimator is explicitly a SimpleBaseline-derived ResNet18 full-image pallet9 adaptation, not PVNet, PoseFix or a renamed DOPE stage. The ImageNet file is `resnet18-f37072fd.pth`, with full SHA-256 recorded in the baseline protocol. All learned body tensors and running statistics must match. The official legacy file lacks all 20 BN batch counters; the loader initializes only that complete missing counter set to zero, as in torchvision’s legacy migration. Partial missing counters or missing learned tensors/running statistics stop loading. The three deconvolution layers and nine-channel head receive fresh initialization.

The 384×512 RGB input uses existing source reflect100 or one native-real reflection, aspect fit and centered BGR128 letterbox, followed by ImageNet FP32 normalization. No predicted or GT box is used to crop the baseline input. Actual per-axis rounded scales and inverse matrices are retained. Nine 96×128 Gaussian heatmaps use continuous point/4 centers, sigma 2 and square 3-sigma truncation. Source visibility greater than zero, finite non-sentinel coordinates and in-grid centers define supervision. Invalid channels remain in the fixed-nine denominator; all-masked frames remain in the batch denominator. The per-frame visibility, finite, outside-support, supervised and all-masked counts are recorded, rather than treating a low loss from absent targets as accuracy.

The specified 60-epoch baseline performs 3498 full 16-image updates plus one 12-image update each epoch, giving 3499 updates per epoch and 209940 total. The source order is a deterministic per-epoch permutation; physical batch 16 is the specified default, without a silent memory fallback. Brightness, contrast and saturation factors are independently drawn per image from $[0.8, 1.2]$ in that fixed order. There is no geometric augmentation. Model and target arithmetic are FP32 with AMP and TF32 disabled. Adam uses betas $(0.9, 0.999)$, learning rate 0.001, weight decay 0.0001, with drops at the beginning of epochs 40 and 50. No real score chooses the schedule, threshold, stopping epoch or checkpoint.

Epoch loss is weighted by actual frame counts, including the final 12-frame batch. Training supervision counts cover 60 epochs of exposures rather than unique images. The calibration 1004 loss and full mask counts are recorded without PnP, heldout or real accuracy selection. Checkpoints carry the in-progress epoch numerator/frame count and cumulative mask counts, so an infrastructure resume does not double-count them. Private DataLoader generators isolate the saved model/augmentation RNG stream; final export can be completed after interruption without replacing a verified final model. The prescribed eight-update GPU smoke is discarded and excluded from the main budget; no smoke speed or learned accuracy is reported here.

After the independently trained baseline is frozen, its layer2 and layer3 outputs give 128 channels at stride 8 and 256 at stride 16. New P/D heads use the unchanged local evidence and source-selection principles for three seeds and 6000 updates of 16 examples per head. The trained baseline checkpoint, baseline protocol and baseline-completion SHA are required before that second protocol can be created. An ImageNet-only network is never silently used as the trained pallet baseline. The planned ResNet18 baseline is 60 epochs of 55,980 source images: 209,940 updates and 3,358,800 exposures, retaining the final 12-image batch. Adam starts at 10^{-3} with weight decay 10^{-4} and factor-ten drops at the start of epochs 40 and 50. Only calibration loss is monitored and final epoch 60 is fixed. Seed 42 and source-only photometric augmentation are prescribed. These are preparation details; actual training and outcomes have not been inserted.

15 ResNet18 results and cost

ResNet18 per-seed table: pending. Seven rows (ResNet18, D1–D3, P1–P3) are reserved; their complete denominator is 319 images and 2818 supervised landmarks. ResNet18 paired intervals: pending. The frozen session-bootstrap

outputs will be inserted without dropping support mismatches or failed resamples. ResNet18 source evidence: pending. Actual usable rows, head parameter counts, completed updates, calibration temperatures, selected movement rules and heldout results require their own receipts. ResNet18 baseline completion and its full 60-epoch training/calibration loss curve remain pending. No simulated curve is shown. ResNet18 cost measurements: pending. Historical YOLO latency is not an estimate of this path. The prescribed seed-1 timing panel must finish before reporting ResNet18 overhead. These results reuse DEV and the geometry-derived reference; completion of three fixed applications does not create independent TEST or a physical T/R stability guarantee. The ResNet18 insertion record is `RESNET_RESULTS_BINDINGS.json`.

16 Reproduction and access

From the repository root, use the recorded environment and `run.py status`, `run.py all`, or `run.py resume` under `scripts/research/pallet_sensors_submission_v1/`. Exact shell commands, downloads, locks and stage receipts are in `REPRODUCE.md`. Completion receipts are checked against actual input/output hashes, not accepted solely on a status string. Main uses no branch rewrite, force push or deletion of unrelated user changes.

Public materials contain code, small numerical results, plots and manuscript sources. Raw training/evaluation imagery, video, feature caches, model weights and the isolated environment remain excluded from public Git. Access restrictions and licenses must be reviewed before any separate sharing. No unverified speedup, annotation-cost reduction, or public-data right is inferred.